23A01604b • OPEN CHANNEL FLOW

UNIT – 4

Sediment Transport

Complete Study Notes • Concepts · Derivations · Formulae · Exam Questions

Syllabus Topics

- Sediment Properties
- Inception of Sediment Motion
- Bed Forms
- Bed Load, Suspended Load and Total Sediment Transport
- Design of Stable Channels and Regime Channels
- Reservoir Sedimentation and Trap Efficiency

1. INTRODUCTION TO SEDIMENT TRANSPORT

1.1 Why study sediment?

Rivers and canals carry water and sediment together. Water flowing over an erodible boundary can scour the bed and banks, carry the material, and deposit it elsewhere. Sediment transport governs river morphology, silting of canals and reservoirs, scour around bridge piers, stability of channels and the life of dams.

1.2 Sources and classification of sediment

- **By origin:** erosion from catchment (sheet, rill, gully), bank erosion, bed erosion; wash load comes mostly from the catchment.
- **By mode of transport:** (a) **bed load** – rolling, sliding, saltating near the bed; (b) **suspended load** – supported by turbulence; (c) **wash load** – very fine particles (< 0.0625 mm) that stay in suspension and are not found in the bed material.
- **By source: bed-material load** = bed load + the suspended part of bed material; it depends on the flow and can be predicted. **Wash load** depends on supply from the catchment, not on the transport capacity.

★ **Key Point:** Total load = bed load + suspended load = bed-material load + wash load. Typically, bed load is 5–25 % of the total load in sand-bed rivers.

1.3 Types of sediment-transport problems

- Alluvial channels (bed and banks are made of the transported material): design of stable canals, regime channels.
- Aggradation and degradation of the bed (silting upstream of weirs, scouring downstream of dams).
- Local scour (bridge piers, abutments, aprons).
- Reservoir and canal sedimentation.

2. SEDIMENT PROPERTIES

2.1 Particle size

Size is the most important property. Defined by the **sieve diameter** (sand and gravel) or **sedimentation diameter** (silt and clay). Grain size classes (modified Wentworth scale):

Class	Size range (mm)	Remark
Boulder	> 256	Moves only in extreme floods
Cobble	64 – 256	Mountain streams
Gravel	2 – 64	Moves as bed load
Sand	0.0625 – 2	Bed + suspended load
Silt	0.004 – 0.0625	Mainly suspended
Clay	< 0.004	Cohesive; wash load

Characteristic sizes: d_{50} (median; 50 % finer by weight), d_{35}, d_{65}, d_{84}, d_{90}. **Geometric mean** $d_g = \sqrt{(d_{84}\, d_{16})}$; **gradation coefficient** $\sigma_g = \sqrt{(d_{84}/d_{16})}$ ($\sigma_g < 1.3$: uniform; > 3: poorly sorted). Sieve analysis gives the **grain-size distribution curve** (percent finer vs log d).

2.2 Shape and density

- **Shape factor** SF = c/√(ab), where a, b, c are the longest, intermediate and shortest axes (natural sand ≈ 0.7). **Sphericity, roundness.**
- **Specific gravity** $s = \rho_s/\rho \approx 2.65$ for quartz; submerged specific weight $\gamma' = (s - 1)\gamma = 16.2$ kN/m^3.
- **Porosity** $p = V_{void}/V_{total} \approx 0.35$–$0.45$ for sand; **dry bulk density** $= (1 - p)\,\rho_s$; dry density of deposited sediment (reservoir): sand 1500–1700 kg/m^3, silt 1000–1400, clay 500–900.
- **Angle of repose φ:** 30°–40° for sand; 35°–45° for gravel; increases with grain size and angularity.

2.3 Fall (settling) velocity

Velocity at which a particle settles in still water, when drag = submerged weight. Drag coefficient C_D depends on particle Reynolds number $Re_p = w\,d/\nu$.

Stokes ($d < 0.1$ mm, $Re_p < 1$): $w = (s - 1)\, g\, d^2 / (18\, \nu)$

Rubey (general, natural sand): $w = F\sqrt{[(s - 1)\, g\, d]}$, $F = \sqrt{(2/3 + 36\nu^2/((s - 1) g\, d^3))} - \sqrt{(36\nu^2/((s - 1) g\, d^3))}$

Large grains ($Re_p > 1000$): $w = 1.1\sqrt{[(s - 1)\, g\, d]}$. Typical values for quartz in water at 20 °C: d = 0.1 mm → 0.009 m/s; 0.5 mm → 0.07 m/s; 1 mm → 0.12 m/s; 10 mm → 0.44 m/s. Fall velocity is reduced by high concentration (hindered settling), turbulence, and increased by flocculation of fines.

2.4 Dimensionless parameters

Particle Reynolds (shear) number: $Re_* = u_*\, d / \nu$

Dimensionless grain diameter: $D_* = d\,[(s - 1)\, g / \nu^2]^{1/3}$

Shields (mobility) parameter: $\tau_* = \tau_0 / [(\gamma_s - \gamma)\, d]$

Here **shear velocity** $u_* = \sqrt{(\tau_0/\rho)} = \sqrt{(g\ R\ S_0)}$ for uniform flow. $Re_* < 5$: hydraulically smooth; $Re_* > 70$: fully rough bed.

3. INCEPTION OF SEDIMENT MOTION

3.1 Forces on a bed particle

A particle resting on the bed experiences the **drag force** F_D and **lift force** F_L from the flow, and its **submerged weight** W' resisting movement through friction at contact points (tan φ). Motion begins when the moment of the driving forces about a contact point exceeds the moment of the resisting forces. Because turbulence makes the instantaneous forces fluctuate, the "critical" condition is statistical.

3.2 Critical shear stress – Shields criterion

Dimensional analysis shows that at incipient motion

$$\tau_{*c} = \tau_c / [(\gamma_s - \gamma)\, d] = f(Re_*)$$

Shields diagram is the plot of τ_{*c} against Re_*. Its main features:

Region	Re_*	τ_{*c}	Remark
Hydraulically smooth	< 5	≈ 0.1 and falling as Re_* rises	Particle buried in viscous sublayer
Transition	5 – 70	min ≈ 0.03 at Re_* ≈ 10	Lowest τ_{*c}
Fully rough	> 70 (> 400)	constant ≈ 0.045 – 0.06	Typical gravel and coarse sand

So for coarse particles: $\tau_c = 0.045\,(\gamma_s - \gamma)\, d$. For quartz in water: $\tau_c \approx 0.045 \times 16\,200 \times d = 730\, d$ (Pa, d in m) → $\tau_c \approx 0.73\, d$(mm) Pa. For d = 1 mm, τ_c ≈ 0.5–0.8 Pa.

Critical velocity: from $u_{*c} = \sqrt{(\tau_c/\rho)}$ and a logarithmic velocity profile (or Manning) the mean critical velocity can be derived; empirical relations (Neill, Garde) have the form $V_c = K\, y^{1/6}\, d^{1/3}$ with K ≈ 6 (SI units).

3.3 Procedure to check incipient motion (design)

- 1. Compute $\tau_0 = \gamma\, R\, S$ (or $\gamma\, y\, S$ for wide channel).
- 2. Compute $Re_* = u_* d/\nu$ and read τ_{*c} from Shields diagram (or use 0.045).
- 3. Compute $\tau_c = \tau_{*c}(\gamma_s - \gamma)d$.
- 4. If $\tau_0 > \tau_c$ the bed moves; otherwise stable. Ratio τ_0/τ_c = **excess shear**.

3.4 Effect of bed slope and side slope

On a bank of slope angle θ (θ < φ) the critical shear stress is reduced by factor K:

$$K = \tau_{c,side} / \tau_{c,bed} = \sqrt{(1 - \sin^2\theta / \sin^2\varphi)} = \cos\theta\, \sqrt{(1 - \tan^2\theta/\tan^2\varphi)}$$

Example: φ = 35°, side slope 1.5H : 1V (θ = 33.7°): $K = \sqrt{(1 - (0.555/0.574)^2)} = 0.25$, i.e. bank particles are much less stable.

3.5 Other criteria

- **White (1940):** turbulence-based, includes hiding effect.
- **Yalin:** τ_{*c} as function of D_*.
- **Shields formula for gradation:** use d_{50} for uniform sand; for mixtures, finer fraction hide behind coarse ones (**hiding factor**), and coarse fractions are more exposed. Armouring occurs when finer fractions are winnowed away, leaving a coarse protective layer (e.g., d_{90} layer).
- **Permissible (Lane/Fortier-Scobey) velocity** used in practice for canals (see Section 8).

✎ **Exam Question:** Explain Shields' criterion for initiation of sediment motion. Determine whether a sand bed with d = 2 mm will move in a wide channel of depth 1.2 m and slope 0.0005. *Ans:* $\tau_0 = 9810 \times 1.2 \times 0.0005 = 5.9$ *Pa.* $\tau_c = 0.045 \times 16\,200 \times 0.002 = 1.46$ *Pa.* $\tau_0 > \tau_c$ *(4× larger), so the bed material moves.*

4. BED FORMS

4.1 Why bed forms form

In an alluvial channel with sand beds, the bed does not remain flat once sediment begins moving. The flow interacts with the bed and creates regular patterns called **bed forms**. They change flow resistance (Manning n and friction factor), sediment transport rate and water level, so they matter in design and in flood stage prediction.

4.2 Types of bed forms (flow regime concept – Simons & Richardson)

Regime	Bed form	Fr	Water surface relation	Resistance / n
Lower regime	Ripples (d < 0.6 mm), ripples on dunes, dunes	< 1 (Fr 0.1–0.6)	Out of phase with bed	High: n ≈ 0.018–0.035
Transition	Washed-out dunes / undulating bed	≈ 0.6–1	Unstable	Falls sharply with V
Upper regime	Plane bed (flat)	0.8–1.2	—	Low: n ≈ 0.012–0.016
Upper regime	Antidunes (standing waves)	> 1	In phase with bed	Low
Upper regime	Chutes and pools	> 1.5	Rapids and jumps	Moderate–high

- **Ripples:** small (wavelength < 0.6 m, height < 0.06 m) triangular forms with a gentle upstream slope and steep lee face; they appear only for fine sand (d < 0.6 mm); they migrate downstream.
- **Dunes:** larger (wavelength 0.6–1000 m, height 0.06–10+ m), out-of-phase with the water surface. Sediment moves up the stoss side, avalanches down the lee, and so the dune moves downstream; flow separation behind the crest produces form drag. Height may be up to y/6; wavelength 5y.
- **Flat (plane) bed:** at higher velocity, dunes wash out; skin friction only; resistance is low, transport is high.
- **Antidunes:** surface and bed waves in phase; waves may move upstream, downstream or stand still; breaking waves occur.

4.3 Resistance to flow – skin and form friction

Einstein & Barbarossa: total resistance R = R' + R'' (R' = grain/skin; R'' = form). Only the grain-resistance part contributes to sediment transport. Strickler: $n' = d_{50}^{1/6}/21$ (d in m) or $n' = d_{90}^{1/6}/26$.

★ **Key Point:** In the lower regime the water surface level rises (resistance increases) when the discharge increases only slightly; in the transition the stage may drop with increasing discharge (discontinuous rating curve) because dunes wash out.

4.4 Prediction of bed-form type

Use diagrams: Simons–Richardson (stream power vs d_{50}), Liu, Van Rijn (D_* vs transport stage parameter $T = (\tau'_b - \tau_c)/\tau_c$: roughly ripples for T < 3, dunes for 3 < T < 15, plane bed for T > 15), Engelund–Hansen.

5. BED LOAD TRANSPORT

5.1 Definition

Bed load is the sediment that moves in continuous or intermittent contact with the bed by rolling, sliding and hopping (saltation). It is expressed as q_b (mass or weight or volume per unit width per unit time). The bed-load layer is only a few grain diameters thick.

5.2 Du Boys formula (1879)

Sediment moves as layers of thickness proportional to the excess shear stress:

$q_b = \psi\, \tau_0\, (\tau_0 - \tau_c)$ where ψ is the Du Boys characteristic coefficient (depends on grain size, found by Straub from flume data) and τ_c comes from Shields. It is the oldest bed-load formula and is mainly of historical value now.

5.3 Meyer-Peter and Müller (MPM, 1948)

The most widely used formula for gravel and coarse sand (d from 0.4 to 30 mm):

$$\Phi = q_b / \sqrt{((s-1)\, g\, d^3)} = 8\, (\tau_*' - 0.047)^{3/2}$$

where Φ is the **dimensionless bed-load intensity** (q_b is the volumetric rate per unit width) and $\tau_*' = \tau_0'/((\gamma_s - \gamma)\, d)$ is the Shields parameter based on **grain** shear stress $\tau_0' = \gamma\, R'\, S$ (skin friction only). In terms of weight: $g_b = \gamma_s\, q_b$.

Worked example: d = 5 mm gravel, s = 2.65, wide channel depth 1.0 m, S = 0.002, no bedforms ($\tau_0' = \tau_0$). $\tau_0 = 9810 \times 1.0 \times 0.002 = 19.6$ Pa; $\tau_* = 19.6/(16\,200 \times 0.005) = 0.242$. $\Phi = 8(0.242 - 0.047)^{1.5} = 8 \times 0.0861 = 0.689$. $\sqrt{((s-1)g\, d^3)} = \sqrt{(1.65 \times 9.81 \times 1.25\times10^{-7})} = 1.42 \times 10^{-3}$ m^2/s. $q_b = 0.689 \times 1.42\times10^{-3} = 9.8\times10^{-4}$ m^3/s/m ≈ 2.6 kg/s/m (with $\rho_s = 2650$).

5.4 Einstein bed-load function (1950)

A **probabilistic** approach: the probability p that the instantaneous lift force on a grain exceeds its submerged weight is related to the rate of erosion, which is balanced by deposition. The dimensionless transport intensity Φ_* and flow intensity $\Psi = (s - 1)d/(R'S)$ are related by a single curve:

$$1 - (1/\sqrt{\pi}) \int e^{-t^2}\, dt = A_* \Phi_* / (1 + A_* \Phi_*) \quad (\text{limits} \pm (B_* \Psi - 1/\eta_0))$$

with $A_* = 43.5$, $B_* = 0.143$, $\eta_0 = 0.5$. Valid for sand and gravel; includes hiding and lift correction.

5.5 Other formulae

Formula	Form	Remark
Bagnold (1956)	$q_b \propto \omega$ (stream power)	Power approach; efficiency $e_b \approx$ 0.1–0.15
Einstein–Brown	$\Phi = 40\, \tau_*^3$	For $\tau_* > 0.18$
Van Rijn (1984)	$q_b = 0.053\, \sqrt{((s-1)g)}\, d_{50}^{1.5}\, D_*^{-0.3}\, T^{2.1}$	Sand 0.2–2 mm, T = transport stage
Parker, Wilcock–Crowe	Fractional transport	Gravel-bed rivers
Wong–Parker (2006)	$\Phi = 4.93(\tau_* - 0.0495)^{1.6}$	Improved MPM

6. SUSPENDED LOAD TRANSPORT

6.1 Mechanism

When the turbulence intensity (vertical fluctuations ≈ u_*) exceeds the fall velocity w, particles remain in suspension. Upward turbulent diffusion balances downward settling at equilibrium:

$$w\,C + \varepsilon_s\,dC/dy = 0$$

C = volumetric (or mass) concentration at height y above the bed; ε_s = sediment diffusion coefficient.

6.2 Rouse equation (vertical concentration profile)

Assume $\varepsilon_s = \beta\,\varepsilon_m$ with parabolic eddy viscosity $\varepsilon_m = \kappa\,u_*\,y\,(1 - y/h)$ ($\kappa = 0.4$, $\beta \approx 1$). Then integration gives

$$C/C_a = [\,((h - y)/y) \times (a/(h - a))\,]^z,\quad z = w / (\beta\,\kappa\,u_*)$$

C_a = reference concentration at height a above the bed (usually a = 0.05 h or 2d); h = flow depth; z = **Rouse number**.

Rouse number z	Mode of transport
> 2.5	Bed load
1.2 – 2.5	Suspended load (50 % of the flow depth)
0.8 – 1.2	Suspended load (100 % depth)
< 0.8	Wash load

★ **Key Point:** Larger z (heavy particles, low shear) gives a steep concentration profile concentrated near the bed; small z (fine particles, strong turbulence) gives uniform concentration over depth.

6.3 Suspended load rate

$$q_s = \int_a^h C(y)\,u(y)\,dy$$

Using the log-law velocity $u(y) = (u_*/\kappa)\,\ln(30y/k_s)$ and the Rouse profile, Einstein derived $q_s = 11.6\,u_*\,C_a\,a\,[P_E\,I_1 + I_2]$, where I_1, I_2 are Einstein integrals (tabulated) in terms of z and a/h, and $P_E = 2.303\,\log(30.2\,h/\Delta)$.

6.4 Measurement and analysis

- **Point-integrating or depth-integrating samplers** (US D-74, DH-48) collect water–sediment mixture. Concentration in mg/L (ppm). **Sediment discharge:** Q_s (t/day) = 0.0864 × C (mg/L) × Q (m^3/s).
- **Sediment rating curve:** $Q_s = a\,Q^b$ (b ≈ 1.5–3). Used for long-term sediment yield with the flow duration curve.
- Modern methods: turbidity meters, acoustic Doppler (ADCP), laser diffraction (LISST).

✎ **Exam Question:** Derive the Rouse equation for the distribution of suspended sediment concentration. What does the Rouse number indicate? (10 marks)

7. TOTAL SEDIMENT TRANSPORT

7.1 Methods

Total bed-material load ($q_t = q_b + q_s$) can be computed in two ways: (i) **separately** compute bed load and suspended load and add (Einstein, Bagnold, Van Rijn); (ii) **directly** compute the total load using a single formula related to the flow (Engelund–Hansen, Ackers–White, Yang).

7.2 Einstein's procedure (1950)

Combine the bed-load function (Section 5.4) with the suspended-load equation at reference level a = 2d; compute for each size fraction of the bed material, weight by its proportion i_b, and sum. Includes separation of the total hydraulic radius into R' (grain) and R'' (form). Requires iterations and charts. Accurate but laborious.

7.3 Engelund–Hansen (1967) – total load

Based on stream-power concept; used for sand-bed rivers (d > 0.15 mm, Fr < 1) with dunes:

$$g_t = 0.05\,\gamma_s\,V^2\,\sqrt{[\,d_{50}/((s-1)\,g)\,]} \times [\,\tau_0/((\gamma_s-\gamma)\,d_{50})\,]^{3/2}$$

g_t is the total load in weight per unit width per unit time (N/s/m); V is the mean velocity; τ_0 is the total bed shear stress (not corrected for form drag, which is built into the formula). Good for dune and plane-bed regimes.

7.4 Ackers–White (1973) and Yang (1973)

- **Ackers–White:** uses mobility number F_{gr} and dimensionless transport $G_{gr} = C(F_{gr}/A - 1)^m$, with coefficients that depend on D_* (for $D_* > 60$: C = 0.025, m = 1.5, A = 0.17). Calibrated for 0.04–4 mm flume data, Fr < 0.8.
- **Yang's unit stream power:** $\log C_t = 5.435 - 0.286 \log(wd/\nu) - 0.457 \log(u_*/w) + (1.799 - 0.409 \log(wd/\nu) - 0.314 \log(u_*/w)) \log(VS/w - V_{cr}S/w)$. C_t in ppm by weight. The sediment concentration is a function of the dimensionless unit stream power VS/w.
- **Brownlie, Van Rijn, Laursen** – field-validated alternatives.

7.4 Comparison of approaches

Formula	Type	Applies to
MPM	Bed load	Gravel and coarse sand, plane bed
Einstein	Bed + suspended	All sizes; needs bed-material gradation
Engelund–Hansen	Total load	Sand beds with dunes
Ackers–White	Total load	Sand and fine gravel
Yang	Total load (concentration)	Sand and gravel

7.5 Practical considerations

- Prediction accuracy is rarely better than a factor of 2; select a formula tested in conditions similar to your river and validate with measured data.
- Use mean annual sediment load = $\int Q_s(Q)\, f(Q)\, dQ$ with the flow duration curve. Most sediment moves during moderately high floods ("effective discharge", about bankfull flow).
- Sediment budget and continuity (**Exner equation**): $(1 - p)\,\partial z_b/\partial t + \partial q_t/\partial x = 0$. If the transport capacity decreases downstream ($\partial q_t/\partial x < 0$) deposition (**aggradation**) occurs; if it increases, scour (**degradation**).

8. DESIGN OF STABLE CHANNELS

8.1 Types of channels

- **Rigid-boundary (lined) channels** – no scour or deposition: designed by Manning's equation.
- **Erodible but non-silting channels (stable)** – the boundary is erodible but the channel does not change shape; designed by **permissible velocity** or **tractive force** methods.
- **Alluvial channels carrying sediment** – the channel adjusts its slope, width and depth: designed by **regime theory** (Kennedy, Lacey, Blench) or rational (Simons–Albertson) methods.

8.2 Maximum permissible velocity method (Fortier & Scobey)

Select a maximum velocity V_{max} that will not erode the channel bed and banks for the type of material.

Material	V_{max} (m/s), clear water
Fine sand	0.45
Sandy loam	0.55
Silt loam	0.60
Stiff clay	1.15
Fine gravel	0.75
Coarse gravel	1.20
Cobbles	1.50
Hard pan / soft rock	1.8 – 2.0

Procedure: choose n, side slopes m; compute R from $V = (1/n)R^{2/3}S^{1/2}$ with $V = V_{max}$, $A = Q/V$; select a bed width b and depth y that satisfy $A = (b + my)y$ and $R = A/P$.

8.3 Tractive force method (Lane, 1955)

The channel is stable if the shear stress on the bed and the banks is less than the critical shear (Section 3).

- Maximum shear stress on the **bed**: $\tau_b = \gamma\, y\, S_0$ (y = depth)
- Maximum shear stress on the **side slopes**: $\tau_s = 0.76\, \gamma\, y\, S_0$
- Permissible (critical) shear stress for side particle: $\tau_{c,s} = K\, \tau_{c,b}$, K as given in 3.4.

Design conditions:

$\gamma\, y\, S_0 \le \tau_{c,b}$ and $0.76\, \gamma\, y\, S_0 \le K\, \tau_{c,b}$ (so the controlling check is the side: $y \le K\, \tau_{c,b}/(0.76\, \gamma\, S_0)$)

For coarse non-cohesive material Lane's rule is $\tau_c \approx 0.75\, d_{75}$ (N/m^2, d_{75} in mm), consistent with Shields: fine sand ≈ 1 N/m^2; gravel with d = 25 mm ≈ 19 N/m^2.

8.4 Design steps for a trapezoidal stable channel

- 1. Choose side slope m (≥ angle of repose) and Manning n.
- 2. Choose bed shear limit τ_c and side slope reduction factor K.
- 3. Determine S_0 (given) and maximum allowable depth y from the side condition.
- 4. Choose b/y; compute A, R and Q from Manning. Adjust b until Q matches the design discharge.
- 5. Check V, Fr (< 0.8 to avoid instability), freeboard.

9. REGIME CHANNELS – KENNEDY AND LACEY THEORY

9.1 Regime concept

An alluvial channel carrying silt-laden water is in **regime** when, over a long period, the net scour and deposition are zero and the channel keeps constant width, depth and slope. Lacey defined three conditions for **true regime:** (i) channel flows uniformly in unlimited incoherent alluvium of the same character as that transported; (ii) discharge constant; (iii) silt grade and silt charge constant. In practice **initial regime** (bed only adjusted) or **final regime** (width also adjusted) is the best attained.

9.2 Kennedy's theory (1895)

From his observations on Upper Bari Doab canals in Punjab: **non-silting, non-scouring velocity** V_0 depends on depth y:

$$\mathbf{V_0 = 0.55\ m\ y^{0.64}}$$

m = **critical velocity ratio** (CVR) = V/V_0: m = 1 for ideal regime; m > 1 (> 1.1) scour; m < 1 deposition. Design: assume y, compute $V_0 = 0.55\ m\ y^{0.64}$, $A = Q/V_0$, then b from $A = (b + 0.5y)y$ (side slope 0.5H:1V), then S from Kutter's (or Manning's) formula; repeat until the CVR is about 1. It is a trial-and-error method; limitations: the relation takes no account of the silt charge or silt grade; width/depth relation is not obtained from the theory.

9.3 Lacey's theory (1930)

Based on data of stable canals in India and Egypt. Lacey considered the **silt factor f**:

$$\mathbf{f = 1.76\ \sqrt{d_{mm}}\ (d_{mm} = \text{mean grain size in millimetres})}$$

Lacey's regime equations (SI units):

$$\mathbf{V = (Q\ f^2 / 140)^{1/6}\ ;\ R = 5V^2 / (2f)}$$

$$\mathbf{P = 4.75\ \sqrt{Q}\ ;\ S = f^{5/3} / (3340\ Q^{1/6})}$$

and **A = Q/V**, which equals P·R for the regime section.

Other relations: wetted perimeter $P \propto \sqrt{Q}$ (width $\propto \sqrt{Q}$), $R \propto Q^{1/3}$, $S \propto Q^{-1/6}$, $V \propto Q^{1/6}\ f^{1/3}$.

Worked example: Design a regime channel for Q = 40 m^3/s, d = 0.3 mm (Lacey).

- f = 1.76 √0.3 = 0.964.
- $V = (40 \times 0.964^2/140)^{1/6} = (0.2655)^{1/6} = 0.80$ m/s.
- $A = Q/V = 50\ m^2$.
- $R = 5 \times 0.80^2/(2 \times 0.964) = 1.66$ m.
- P = 4.75 √40 = 30.0 m (check $A = P\ R = 49.8\ m^2$ ✓).
- $S = 0.964^{5/3}/(3340 \times 40^{1/6}) = 0.941/6176 = 1.52 \times 10^{-4} \approx 1$ in 6560.

9.4 Comparison of Kennedy and Lacey

Feature	Kennedy	Lacey
Basis	Silt-supporting eddies from the bed only	Eddies from both bed and sides; regime concept
Silt grade	Through CVR m only	Silt factor f
Section	Any width/depth, trial-and-error	Unique section for Q and f
Section shape	Trapezoidal	Semi-elliptical (regime) in theory
Equations	One (V_0 = 0.55 m $y^{0.64}$) + Kutter	Three main regime equations
Accuracy	Lower	Better (limited by data)

9.5 Blench and Simons–Albertson

- **Blench (1952):** introduces a bed factor F_b and a side factor F_s ($F_s \approx 0.1$ for slightly cohesive, 0.2 for medium, 0.3 for highly cohesive banks): width $b = \sqrt{(F_b Q/F_s)}$; depth $d = (F_s Q/F_b^2)^{1/3}$. Accounts for bank strength and sediment concentration.
- **Simons & Albertson (1963):** classify channel into 5 types by bed and bank material and sediment load; gives regime relations: $P = K_1 Q^{0.512}$, $R = K_2 Q^{0.361}$ etc. Based on data from Indian and American canals.
- **Modern rational (analytical) regime methods:** White–Bettess–Paris, Parker's gravel-bed river theory, extremal hypotheses (minimum stream power, maximum transport).

✎ **Exam Question:** Design an irrigation channel by Lacey's theory for Q = 30 m^3/s, silt factor 1.0. *Ans: $V = (30\times1/140)^{1/6} = 0.774$ m/s; $A = 38.8\ m^2$; $R = 5\times0.599/2 = 1.50$ m; $P = 4.75\sqrt{30} = 26.0$ m; $S = 1/(3340\times30^{1/6}) = 1/5885 = 1.7\times10^{-4}$.*

10. RESERVOIR SEDIMENTATION AND TRAP EFFICIENCY

10.1 The problem

A dam interrupts the river flow. The reduced velocity in the reservoir causes the coarse fractions to settle at the head (**delta**) and fine fractions to settle further in (**bottom sets**) or be carried as **turbidity (density) currents** along the old river bed to the dam. Deposition occupies storage capacity: dead, live and flood-control storage, reduces energy and water supply, raises upstream flood levels (backwater), damages turbines and gates (abrasion) and starves the downstream river of sediment (degradation).

Worldwide, reservoirs lose about 0.5–1 % of storage per year; in India the rate is 0.5–1.5 % and in the Himalayan region more.

10.2 Sediment yield from the catchment

- **Measured:** sediment rating curve + flow duration curve; reservoir surveys (hydrographic, capacity surveys with echo-sounders).
- **Empirical: USLE** A = R K L S C P (soil loss in t/ha/yr; R rainfall erosivity, K soil erodibility, LS slope length and steepness, C cover, P practice); **MUSLE** for single storms; sediment delivery ratio (SDR) accounts for the proportion reaching the outlet; **Khosla's and Joglekar's** empirical formulae for Indian catchments relate annual sediment yield to catchment area.
- Typical yields: 50–500 t/km^2/yr (temperate) to > 2000 t/km^2/yr (Himalayan, loess).

10.3 Trap efficiency

Trap efficiency (TE) is the proportion of the incoming sediment that is deposited in the reservoir:

$$\text{TE} = (\text{sediment inflow} - \text{sediment outflow})/\text{sediment inflow} \times 100\ \%$$

Depends on: capacity/inflow ratio, size and settling velocity of particles, reservoir shape, operation (flushing, sluicing), detention time.

Brune's curve (1953): empirical relation between TE and the capacity–inflow ratio C/I (C = reservoir capacity, I = mean annual inflow volume), derived from 40 normal-ponded reservoirs. Approximate values from the median curve:

C/I	TE (approx.)
1.0	≈ 97 %
0.1	≈ 90 %
0.01	≈ 55–70 %
0.001	≈ 10–30 %

Brune's curves: upper envelope – coarse sediment; median – normal; lower envelope – fine, colloidal sediments.

Churchill (1948): TE is a function of the **sedimentation index** SI = (retention period)/(mean velocity through the reservoir); TE falls as SI decreases, i.e. as water passes through faster. Mainly for small, short-detention reservoirs.

Camp (1945) / Camp–Dobbins (settling basin) concept: an ideal settling basin removes all particles with $w \geq Q/A_s$ (overflow rate = fall velocity). The fraction removed of each class = $w/(Q/A_s)$ for $w < Q/A_s$. TE of whole reservoir = Σ(size fraction × removal ratio).

10.4 Estimating the reservoir life

Annual sediment volume deposited $V_s = (TE \times W_s)/\rho_{deposit}$, with W_s = annual sediment load (t/yr) and $\rho_{deposit}$ = dry density of deposit (t/m^3; sand 1.5; silt 1.2; clay 0.7 – 1.0; or from **Lara–Pemberton** formula for compaction with time).

Example: capacity 100 Mm^3; sediment inflow 1.5 Mt/yr; TE = 90 %; deposit density = 1.2 t/m^3. V_s = 0.9 × 1.5/1.2 = 1.125 Mm^3/yr, so the **half-life (50 % filled)** ≈ 44 years and complete filling ≈ 89 years (without allowing for decreasing TE as capacity drops).

10.6 Control and management of reservoir sedimentation

Category	Measures
Reduce sediment inflow	Catchment treatment (afforestation, contour bunds, terracing, check dams), sediment-trap dams, vegetative filter strips
Route sediment past the reservoir	Bypass channel or tunnel, off-stream reservoirs, **sluicing** (open gates during floods)
Pass sediment through	**Venting of turbidity currents** through low-level outlets
Remove deposited sediment	**Hydraulic flushing** (draw-down, needs a large low-level outlet), dredging (hydraulic/mechanical), dry excavation
Adaptive strategies	Raise the dam; increase storage upstream; design with extra dead storage (e.g. 50-year sediment allowance); sediment-resistant intakes; trap-efficiency monitoring

★ **Key Point:** Always include a sediment-storage allowance (usually for 50–100 years) in a reservoir's design, and place the power intake and outlets above the expected sediment level.

10.7 Quick revision – formulae

Topic	Formula
Stokes' fall velocity	$w = (s-1)g\,d^2/18\nu$
Shields parameter	$\tau_* = \tau_0/[(\gamma_s - \gamma)d]$; $\tau_{*c} \approx 0.045$
Shear velocity	$u_* = \sqrt{(gRS)}$
MPM	$\Phi = 8(\tau_*' - 0.047)^{3/2}$
Rouse equation	$C/C_a = [(h-y)a/(y(h-a))]^Z$, $z = w/\kappa u_*$
Engelund–Hansen	$g_t = 0.05\gamma_s V^2\sqrt{(d/(s-1)g)}\,\tau_*^{3/2}$
Lacey	$V = (Qf^2/140)^{1/6}$; $R = 5V^2/2f$; $P = 4.75\sqrt{Q}$; $f = 1.76\sqrt{d_{mm}}$
Kennedy	$V_0 = 0.55\ m\ y^{0.64}$
Trap efficiency	Brune curve: TE = f(C/I)

✎ **Exam Question:** What is trap efficiency of a reservoir? Explain the factors influencing it and describe methods to control reservoir sedimentation. (10 marks)